

Vector: addition

Advanced: Practice 2  

Vector ***a*** is defined as (11, -2, -4)

Vector ***b*** is defined as (4, 6, -3)

What is ***a*** + ***b***?

☐ (44, -12, 12)

☐ (15, 8, 7)

☐ (15, 4, -7)

☐ 44

Quiz:

**STEM SMART 2026 Residential Extend Quiz 3 –
Vectors**

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Vector: multiplication

Advanced: Practice 2  

Vector ***a*** is defined as (11, -2, -4)

A scalar ***b*** has the value 12.

What is ***a*** x ***b***?

- ☐ 60
- ☐ (23, 10, 8)
- ☐ (23, 14, 16)
- ☐ (132, -24, -48)

Quiz:

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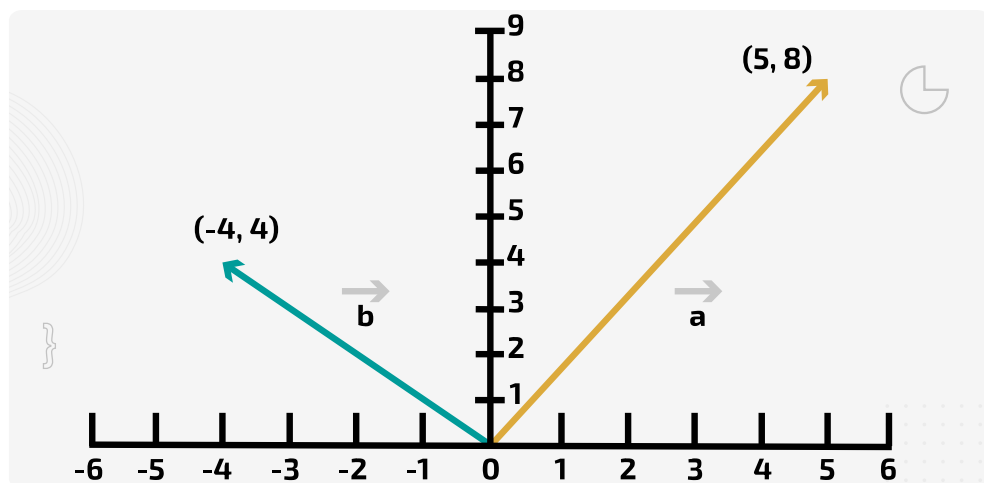
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Vector: calculate dot product

Advanced: Practice 2 ●●

What is the dot product of the vectors $\mathbf{a}$ and $\mathbf{b}$?



Vectors $\mathbf{a}$ and $\mathbf{b}$

Quiz:

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Vector: convex combination

Advanced: Practice 2  

Vector ***a*** is defined as (3, 8)

Vector ***b*** is defined as (9, 3).

If α is 0.7 and β is 0.3, what is the convex combination of ***a*** and ***b***?

☐ (18.9, 17.2)

☐ (4.8, 6.5)

☐ (7.7, 3.6)

☐ (8.4, 3.3)

Quiz:

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Vector: operations

Advanced: Challenge 2 

There are many operations that can be performed on vectors and each operation achieves something different. Which of the following statements relating to vector operations is **not** accurate?

- ☐ Adding two vectors results in displacement or translation.
- ☐ The dot product of two vectors can be used to find the angle between the two vectors.
- ☐ When calculating a convex combination, the sum of α and β must be 0.
- ☐ Scalar-vector multiplication has the effect of scaling a vector.

Quiz:

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